

Exam Date & Time: 31-Dec-2024 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

**Manipal Institute of Technology, Bengaluru**  
BTech VII Semester - Make up Examination - December -2024  
**Data Structures and Algorithms [ECE-4070]**

**Marks: 50**

**Duration: 180 mins.**

### PART-A

**Answer all the questions.**

- 1) Differentiate between Program and Algorithm. Discuss the specifications of an algorithm. (4)
  - A)
  - B) Differentiate between Space complexity and Time complexity. (3)
  - C) Represent and discuss the significance of Sparse matrix transpose. (3)
- 2) Illustrate and explain the postfix expression evaluation in stacks. (4)
  - A)
  - B) Discuss the implementation of Queues using arrays. (3)
  - C) With an example code, explain the deletion operation at the given position for a Singly linked list. (3)
- 3) Illustrate and explain the methods for deletion of a node in case of a Binary Search tree. (4)
  - A)
  - B) Enlist tree traversal operations and explain any one with suitable example. (3)
  - C) Elaborate the terminologies used in AVL tree with proper definitions. (3)
- 4) With an illustration, explain the Incidence matrix to represent graphs. (4)
  - A)
  - B) Suggest an algorithm for Depth First Search(DFS) method. (3)
  - C) Discuss the steps of Insertion sort algorithm. (3)
- 5) Suggest an algorithm for Quick sort highlighting the significance of the same. (4)

- A)
- B) Discuss the steps involved in heap and insertion sort. (3)
- C) Discuss the concept of Breadth First Search(BFS) algorithm. (3)

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